Appendix I - Error Message Index

The printable errors a running Intuition Engine produces. Grouped by source. Each entry has the exact text the machine prints, the numeric code the runtime uses internally, and a one-line explanation.

I.1 IE64 BASIC runtime errors

The format on the screen is the message text, a space, the literal word `ERROR`, then (when raised from a running program rather than from direct mode) `IN` followed by the line number.

Code	Printed text	Meaning
1	`SYNTAX`	The tokeniser or parser cannot make sense of the statement.
2	`DIVISION BY ZERO`	A `/`, `\`, or `MOD` operator saw a zero denominator.
3	`UNDEFINED LINE`	`GOTO`, `GOSUB`, `THEN`, or `RESTORE` referenced a line that does not exist.
4	`NEXT WITHOUT FOR`	`NEXT` did not match a pending `FOR`.
5	`RETURN WITHOUT GOSUB`	`RETURN` did not match a pending `GOSUB`.
6	`OUT OF MEMORY`	The program, variable, array, or string area is full.
7	`ILLEGAL QUANTITY`	A function received an argument outside its domain (negative `SQR`, non-positive `LOG`, out-of-range `CHR$`).
8	`OVERFLOW`	An arithmetic operation overflowed the double-precision numeric range.
9	`TYPE MISMATCH`	A numeric expression saw a string, or vice versa.
10	`FC`	Illegal function call. Raised by `POKE16`, `POKE32`, or `POKE64` to an unaligned address, by a failed `HOST` action, and by other built-in helpers when they reject their argument shape.
11	`REDIM`	`DIM` named an array that already exists.

Additional message strings produced by specific verbs:

Verb context	Printed text
`LOAD`	`?FILE NOT FOUND`.
`SAVE`	`?FILE ERROR` (printed as a soft error; `SAVE` does not raise into the runtime).
`RUN AOT`	`Compiling to native code...` before compilation begins.
`RUN AOT` / `COMPILE` / `TRANSPILE`	`?NO CODE TO COMPILE` when the stored programme is empty.
`RUN AOT` / `COMPILE` / `TRANSPILE`	`?COMPILE ERROR IN <line>: <reason>` when a stored line cannot become native IE64 code.
`COMPILE` / `TRANSPILE`	`?COMPILE ERROR IN <line>: <reason>` when a stored `LOAD` is used. `LOAD` is available to the retained `RUN AOT` target, not to a standalone image.

Verb context	Printed text
RUN AOT / COMPILE / TRANSPILE / ASSEMBLE	?OUT OF MEMORY ERROR IN <line-or-0>: <reason> when the native-code arena, generated source, or output image is too large.
COMPILE / TRANSPILE / ASSEMBLE	?FC ERROR IN 0 for a bad output name.
COMPILE / TRANSPILE	?FILE ERROR IN 0 when the output image or generated source cannot be written.
ASSEMBLE	?FILE ERROR IN 0 when the matching assembly source is missing, unreadable, or too large.
ASSEMBLE	?COMPILE ERROR IN 0 when the IE64 source cannot be assembled.
TYPE	?SYNTAX ERROR IN 0 when the quoted filename is missing or malformed.
TYPE	?FILE NOT FOUND, ?FILE TOO LARGE, or ?FILE ERROR when the text file cannot be read.
TYPE	?NOT A TEXT FILE when the file contains binary control bytes.
RUN AOT / COMPILE / TRANSPILE	?COMPILE ERROR IN <line>: TYPE is direct-only when a stored line tries to compile TYPE.

I.2 Machine monitor (IE Mon)

The monitor prints short, lowercase-prefixed messages. Every unrecognised command prints unknown command. The common ones:

Printed text	Meaning
?	Generic syntax error in a monitor command line.
unknown command	The first token did not name a command.
bad address	An address argument was outside the addressable range.
bad range	An end address was lower than its start address.
bad value	A value argument did not parse as a number.
no breakpoint at <addr>	bc was given an address with no breakpoint.
breakpoint list full	The breakpoint table cannot hold another entry.
cpu frozen	An operation tried to step a frozen CPU; thaw it first.
file error	A save or load sidecar command failed.

The monitor's h (hunt) and c (compare) commands print addresses of matches rather than diagnostic messages. The disassembler prints ??? for an unknown opcode.

I.3 File I/O block

When the File I/O block (Chapter 35) fails, FILE_STATUS reads 1 and FILE_ERROR_CODE is one of:

Code	Meaning
0	OK (paired with FILE_STATUS = 0).
1	Not found.

Code	Meaning
2	Permission.
3	Path traversal.
4	Range error: the staged data span would reach `$FFFF0000`, wrap the `32`-bit pointer, or exceed active RAM.

For a read whose name passes path validation but whose file cannot be opened, `FILE_RESULT_LEN` is cleared to `0`. A program should still test `FILE_STATUS` first and then read `FILE_ERROR_CODE`. The same `0` length is reported when a read or directory listing is refused with range error `4`.

I.4 HOST appliance block

The status byte at `$F1408` (Chapter 36) reads one of:

Code	Meaning
0	Running.
1	OK (terminal).
2	Error (terminal).
3	Cancelled by user (terminal).
4	Disabled (the system-action bridge is off).
5	Idle (no command has been fired).

A non-zero exit code at `$F140C` after a terminal status of `2` gives the underlying action's exit value; the meaning is action-specific and is not normalised across subverbs.

I.5 Shared network sockets

The shared socket block in Chapter 39 reports:

Register	Value	Meaning
HOST_SOCKET_STATUS	0	Ready or last command succeeded.
HOST_SOCKET_STATUS	1	Busy, reserved by the interface.
HOST_SOCKET_STATUS	2	Last command failed.
HOST_SOCKET_RES1	$FFFFFFFF	Integer error result.

Common `HOST_SOCKET_ERRNO` values are:

Code	Meaning
0	No socket error.
9	Bad or unknown descriptor.
22	Invalid argument, descriptor, pointer, or length.
35	Operation would block or released key is unavailable.
40	Message or requested transfer is too large.

Code	Meaning
45	Operation is not supported.
55	No buffer or descriptor capacity remains.
78	Network service unavailable.

HOST_SOCKET_HERRNO value 1 means host not found. Value 3 is defined as a non-recoverable resolver failure; the current resolver path reports its failures as 1.

I.6 RUN loader block

The RUN loader block (RUN "<name>", Chapter 35) reports status:

Code	Meaning
0	Idle.
1	Loading.
2	Running.
3	Error.

On error, the error register reports:

Code	Meaning
0	OK.
1	Not found.
2	Unsupported type.
3	Path invalid.
4	Load failed.

RUN translates a non-zero result into ?FILE ERROR for the file-error cases and ?FC ERROR for the unsupported cases. For .ie64 images, load failed includes an image too large to fit at PROG_START; the image is rejected before it can partially overwrite memory.

I.7 Media loader

The media loader (SOUND PLAY, Chapter 23) reports status:

Code	Meaning
0	Idle.
1	Loading.
2	Playing.
3	Error.

On error, MEDIA_ERROR reports:

Code	Meaning
`0`	OK.
`1`	File not found.
`2`	Bad format or read failure.
`3`	Unsupported extension.
`4`	Invalid filename.
`5`	File too large for the staging buffer.

For MIDI/MUS, bad SMF headers, unsupported SMF type `2`, SMPTE timing, bad MUS score ranges, and unsupported MUS event types all report as bad format.

I.8 Coprocessor

`COSTATUS` (Chapter 32) reports:

Code	Constant	Meaning
`0`	`COPROC_TICKET_PENDING`	Queued, not yet started.
`1`	`COPROC_TICKET_RUNNING`	Worker is processing.
`2`	`COPROC_TICKET_OK`	Completed successfully.
`3`	`COPROC_TICKET_ERROR`	Worker returned an error.
`4`	`COPROC_TICKET_TIMEOUT`	Wait deadline expired.
`5`	`COPROC_TICKET_WORKER_DOWN`	Worker is no longer running.

`COWAIT` blocks until the ticket reaches a terminal state or the timeout expires; call `COSTATUS(ticket)` afterwards to read the final code.

Raw coprocessor commands report their last command result through `COPROC_CMD_STATUS` and `COPROC_CMD_ERROR`:

Code	Constant	Meaning
`0`	`COPROC_ERR_NONE`	No command error.
`1`	`COPROC_ERR_INVALID_CPU`	The selected CPU type is not valid for this command.
`2`	`COPROC_ERR_NOT_FOUND`	The named worker was not found, or `COPROC_CMD_START_MEM` was given an empty or out-of-range guest-RAM image.
`3`	`COPROC_ERR_PATH_INVALID`	The worker name is not an accepted path.
`4`	`COPROC_ERR_LOAD_FAILED`	The worker image could not be loaded or started.
`5`	`COPROC_ERR_QUEUE_FULL`	The request queue cannot accept another entry.
`6`	`COPROC_ERR_NO_WORKER`	The selected worker is not running.
`7`	`COPROC_ERR_STALE_TICKET`	The ticket no longer names a live request.
`8`	`COPROC_ERR_INVALID_INSTANCE`	The CPU type is valid, but the selected instance is beyond its limit.
`9`	`COPROC_ERR_STALE_WORKER`	The worker did not acknowledge the current mailbox layout version.

I.9 Raised by the CPU itself

Per CPU, the chapter (Ch 25-30) lists the trap and exception vectors and their meanings. The monitor's `r` command displays the current trap source when a CPU has stopped at one. Common cross-CPU shapes:

- Division by zero raises the CPU's divide-by-zero exception on M68K (vector `5`) and x86 (`INT 0`). IE32 stops with a division-by-zero error. IE64 integer divide and modulo by zero write `0` and do not trap; IE64 floating-point divide-by-zero is reported in the FPU status register.
- An unaligned `32`-bit access on the IE64 raises an alignment fault; the address that caused it is in `CR_FAULT_ADDR`.
- An undefined opcode raises the CPU's illegal-instruction vector (M68K vector 4, x86 `INT 6`, Z80 silently re-executes on most undocumented prefixes, 6502 documents the undocumented opcodes - see Chapter 27).
- On IE64, `MTCR` to the read-only `CR_RAM_SIZE_BYTES` control register raises `FAULT_ILLEGAL_INSTRUCTION` (cause `11`).